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DEPARTMENT OF
INFORMATICS +
TELECOMMUNICATIONS

Artificial Intelligence II

Homework 4

Bert Models

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All documentation for the project can be found in the two notebooks.

- **Task1_Bert_Classification.ipynb:** This notebook contains the experiment for the Vaccine Sentiment Classifier using a neural network with pretrained bert layer. [source1](#). [source2](#)
- **Task2_QA_Bert.ipynb:** This notebook contains the experiment for the Question/Answering task using pretrained Bert model. we used the DistilBertForQuestionAnswering, due to the limited time available for the model's training. [source1](#) [source2](#).
- **Task3:** This task's implementation and experiments can be found in two notebooks: `squad_model.ipynb` and `triviaqa_model.ipynb`

To sum up, we trained two models that are supposed to be almost the same architecturally (Bert and DistilBert, with two different datasets. There is some vague space there due to the fact that DistilBert is a discounted version of Bert, however, the model obviously fairs better after trained with the squad dataset.

Even during its training evaluation process the train loss and the eval loss relation has a better result by the end of the training in the Bert+Squad case.

Finally, Bert+Squad had f1 score 0.69 for the predictions it made for the questions found in the squad test dataset and came up with 0.54 exact matches, which means that more than half of the questions were answered perfectly.

On the other hand DistilBert+TriviaQa had f1 score 0.31 for the predictions it made for the questions found in the triviaqa test dataset and came up with 0.28 exact matches. These results compared to Bert-Squad case, are disappointing. Especially if we take into consideration that this second model also fails to answer question posed outside of its dataset, while Bert-Squad seems to answer them. [source1](#) [source2](#) [source3](#)